UNIT 5 ENERGY CONSERVATION

What is energy conservation?

- Energy conservation is Utilization of less energy out of available amount without any wastage and affecting productivity and growth rate. Energy conservation can be broadly defined as the efficient use of energy.
- The reduction or elimination of unnecessary or unwanted energy use is referred to as energy conservation.
- It can be accomplished by using less energy to perform a given amount of work or by not using energy at all.
- Energy conservation is the decision and practice of using less energy like:-

1. Turning off the light when you leave the room.
2. Unplugging appliances when they're not in use .
3. Walking instead of driving are all examples of energy conservation.
4. The two main reasons people conserve energy are to gain more control over their energy bill and reduce the demand on the earth's natural resources.

Scope For Energy Conservation

The benefits and scope of energy conservation are:

- It saves money by reducing energy bills.
- It reduces adverse environmental impacts like air pollution, acid rain, global warming water pollution and oil spills.
- It reduces maintenance cost by operating equipment for fewer hours.
- It conserves energy resources.
- It preserves the ecosystem.
- It reduces the shortage of power.

What Is Energy Efficiency?

- Normally efficiency equals to percent ratio of output to input.
- In terms of industries, efficiency means production of more products with the same amount of raw materials.
- Through Energy efficiency we reduce the energy wastage.
- Energy Efficiency increase as energy conservation increase.

Maximum Energy Efficiency:-
When maximum work is done from the given quantity of energy input then the process or system is said to be energy efficient system. The principle of Maximum energy efficiency states that maximum part of energy should be converted into work.

Maximum Cost Effectiveness

- The principle of Maximum Cost effectiveness emphasizes on getting more output in less cost input. So, when a device, system or a process gives maximum production in minimum cost input then the process is said to be cost effective process.
- A system is said to be energy conservative system when both the principle of energy and cost implies.

Methods and techniques of energy conservation in HVAC System

The heating, ventilation and air conditioning system transfer heat energy from the product or to the product.

Components of HVAC's

- Motor
- Pumps
- Fans
- Compressors
- Ducting
- Filters.

1)Methods and techniques of energy conservation in refrigerator

1. **Buy an energy-efficient appliance**
   The information on the energy label gives a clear indication of a refrigerator's energy efficiency. The following rule of thumb applies: the more + signs after the A, the more energy efficient and economical the appliance.
2. **Location is crucial**
   Never place your refrigerator next to a heat source such as an oven, hob or radiator, and avoid direct sunlight. Always ensure that your refrigerator has sufficient ventilation so that the compressor does not end up running continuously.

3. **Always close the door without delay**
   Every time the refrigerator door is opened, cold air escapes and warm ambient air enters. To compensate for the temperature increase in its interior, the refrigerator must then use energy to bring the temperature back down. Always avoid opening the door unnecessarily and for too long.

4. **Only place cold food in the refrigerator**
   Never put warm food in the refrigerator as this will heat up the interior, as well as other stored foods. Hot food should always be allowed to cool to room temperature before placing it in the refrigerator.

5. **Defrost the freezer compartment**
   A thick layer of ice in the freezer compartment acts as a form of insulation and the appliance has to use more energy to cool the interior; therefore, we recommend that you regularly defrost your freezer.!

6. **Check door seals regularly**
   The rubber seal is used for insulation purposes and prevents heat exchange between the ambient air and the refrigerator. It is important to regularly check that the seal is fully intact and sealing tightly. Should this not be the case, replace it immediately! Also, make sure that stored items doesn't prevent the door from closing properly.

7. Get rid of your old refrigerator

2)Methods and techniques of energy conservation in Pump

1. Choose the pump of proper capacity to avoid wastage of electrical energy
2. Controlling the flow rate by speed variation
3. Pumps in parallel to meet varying demand
   Operating two pumps in parallel and turning one of when the demand is lower, can result in significant energy savings. Pumps providing different flow rates can be used.
4. Seals and packaging should be checked properly.
5. Regular maintenance of pump should be done.
6. Start/stop control of pump
7. Operate pumps near their best efficiency point (BEP)
8. Ensure adequate NPSH at site of installation
9. Avoid operating more than one pump for the same application
10. Use booster pumps for small loads requiring higher pressure
11. Repair seals and packing to minimize water loss by dripping
12. Conduct a water balance to minimize water consumption, thus optimum pump operation
13. In multiple pump operations, carefully combine the operation of pumps to avoid throttling.
14. Replace old pumps with energy efficient pumps
15. To improve the efficiency of oversized pumps, install variable speed drive, downsize /replace impeller, or replace with a smaller pump.

3) Methods and techniques of energy conservation in Fans

- While purchasing and installing fans, precautions should be taken to purchase energy efficient fans. They must be proper size of a room, means a fan of 48 cm size is suitable for a room of 10 feet x 10 feet. More than this size will cause unnecessary wastage if energy.
- When installed, we should see that the blades are properly balanced and produce no noise and vibration, which will increase power input.
- Keep proper speed of fans in the rooms.
- Use fans at low speed instead of at high speed.
- Adjust direction of ceiling fan so as to air blow downwards.
- Turn off fans when not required.
- Maintain, repair, use properly in good condition.

- Use the fan which has energy efficient motor.
- Use properly designed blade fans.
- Use electronic controlled regulators instead of conventional regulator
- Use windows to allow natural air circulations.

4) Methods And Techniques Of Energy Conservation In Ovens

- Microwave ovens consume less power as compared to regular conventional oven.
- Due to use of microwave up to 50% electricity can be saved.
- Do not overload the ovens.
- Do not have larger water content directly in oven.
- Use only required amount of water

  A clean oven will heat up quicker and distribute heat more effectively, making it more energy efficient.
- Clean electric ovens regularly.
- Use ovens having automatic shut off button
- Use ceramic or glass dishes

  Glass and ceramic dishes retain heat a lot better than metal dishes, means you can use a slightly lower temperature. A lower temperature means lower energy.

Cook several dishes at once

Without saturating it, one good way to save energy when using your oven is to **cook several dishes at the same time**. Most home ovens have enough space to cook several dishes simultaneously. This way, you get two for the price of one in your kitchen **saving money, energy and time**.

Turn off the oven 5 minutes before

Another way to harness the residual heat is to turn the oven off 5 minutes before finishing cooking your dish. **The residual heat will be enough to finish the dish** without affecting its quality and you could save, using this simple trick.

- **Don't open the door while cooking**

  It is true that often you might have to open the door to check that your dish is turning out as planned. However, by reducing the number of times you open the oven door when cooking, you can avoid a loss of heat, achieving a more efficient and sustainable use.

5) Methods And Techniques of Energy Conservation In Boiler

- Always clean the burners, nozzles and strainers etc
- Preheat combustion air from waste heat.
- Use good quality water in boiler to avoid rusting in boiler equipment.
- Variable speed drives are used for large capacity boilers having variable flow.
- Boiler maintenance should be carryout regularly.
- Use multiple numbers of boilers.
- Steam leakage should be avoided.
- The temperature should change slowly and uniformly.
- The water level in boiler should not allowed to fall beyond the minimum level.
- The boilers should be inspected at least once in year for corrosion, cracks, leaks and other irregularities.

6) Methods And Techniques of Energy Conservation In Lighting Techniques

- Keep lights low or off when not needed
- Turn off lights in any room not being used
- Concentrate lighting in reading areas, work areas, and where lighting is needed for safety
- Install solid-state dimmers
- Turn off outdoor lights when not needed
- Use timers, motion detectors, heat sensors, or photocell lighting controls when possible.
- Utilize sunlight as maximum as possible.
- Use energy efficient lamps e.g. LED, CFL etc.
- Conventional fluorescent lamp should be replaced by energy efficient fluorescent lamp.
- Turn off the lights when not needed.
- Paint the walls with lighter colors because dark colors absorb illumination